Breathing Around the School

Pupil sheet

Name: _____ **Date:** _____

Safety: I stay with the group on the pavement or in the yard, never in the road; I do not hold the box near an exhaust pipe or smoke; I put nothing above the particle sensor.

1. What I see on the screen

The start page shows these sensors in green (tick): ☐ particles (PM2.5, PM10) ☐ NOx index ☐ temperature and humidity

Three rules. I hold the box **at the same height** at every place, lens upwards, **with nothing above it**, in the shade of my body. I choose the place **before** I press **Start recording**. I record **five minutes** at each place, then press **Stop recording**.

2. What I do on my own

First I predict. I do not change my prediction after I measure.
The most particles will be at: ☐ P1 ☐ P2 ☐ P3 ☐ P4
The warmest place will be: ☐ P1 ☐ P2 ☐ P3 ☐ P4

I think so because _____

_____

The four places. I write a typical value from each card (or, in Excel, the mean). Particles in $\mu g/m^3$; for the NOx index I write the largest value I saw.

Place	PM2.5	PM10	NOx index	T (°C)	H (%)
P1 Entrance					
P2 Busy road					
P3 Quiet area					
P4 Green area					

_____

While we measured, I noticed (a bus, wind, smoke, a mower) at _____

_____

The WHO guideline for PM2.5 is 15 $\mu g/m^3$ as a mean over a whole day. Our five minutes at P2 can be compared with it: ☐ yes ☐ no.

3. The conclusion I write

I predicted that, and the data show that _____

_____

The difference between P2 and P4 is larger / smaller than the ups and downs at one place, because _____

_____

The quantity that surprised me was _____

_____

The limit of our measurement: name one thing (wind, time of day, only five minutes) that could change our result on another day.

_____

Exit ticket

Where around our school would you wait for someone to collect you? Give one of our measurements as the reason.

_____